# Acute upper GIT bleeding

## The AIM of this lecture is:

1. To know the definition & causes of upper GIT bleeding.
2. To understand the clinical presentation of acute upper GIT bleeding.
3. To know how to manage acute upper GIT bleeding.

## Intended learning outcomes:

1. To be able to list the possible causes of upper GIT bleeding.
2. To be familiar with the clinical presentation of acute upper GIT bleeding.
3. To know how to diagnose & treat patients with acute upper GIT bleeding

- This is the most common gastrointestinal emergency , Bleeding from the upper GI tract occurs anywhere between the oropharynx and ligament of Treitz which delineates the transition between the duodenum (foregut) and the jejunum (midgut).
- This encompasses the oral cavity, esophagus, stomach (fundus, cardia, body, and pyloric region) as well as the entirety of the duodenum.

- The mortality of patients admitted to hospital is about 10% but there is some evidence that outcome is better when individuals are treated in specialised units.
- Risk scoring systems ( **BLATCHFORD score** ) have been developed to stratify the risk of needing endoscopic therapy or of having a poor outcome.

- The advantage of the Blatchford score is that it may be used before endoscopy to predict the need for intervention to treat bleeding.
- Low scores (2 or less) are associated with a very low risk of adverse outcome.

**Table 1** Glasgow-Blatchford score.

| | GBS |
|---|---|
| **Blood urea (mmol/l)** | |
| <6.5 | 0 |
| 6.5 – 7.9 | 2 |
| 8 – 9.9 | 3 |
| 10 – 25 | 4 |
| >25 | 6 |
| **Hemoglobin: men (g/l)** | |
| >129 | 0 |
| 120 – 129 | 1 |
| 100 – 119 | 3 |
| <100 | 6 |
| **Hemoglobin: women (g/l)** | |
| >119 | 0 |
| 100 – 119 | 1 |
| <100 | 6 |
| **Systolic blood pressure (mmHg)** | |
| >109 | 0 |
| 100 – 109 | 1 |
| 90 – 99 | 2 |
| <90 | 3 |
| **Other markers** | |
| BPM ≥ 100 | 1 |
| Melena | 1 |
| Syncope | 2 |
| Liver disease* | 2 |
| Heart failure* | 2 |

Abbreviations: BPM, beats per minute; GBS, Glasgow-Blatchford Bleeding Score.
Total score is obtained by adding the value of all items.
Adapted from Stanley AJ et al [8].
* Based on past medical history, clinical or laboratory evidence.

## Causes of upper GIT bleeding

1. Peptic ulcer dis. ( H pylori , NSAIDS) ( the most common cause , 35-50% ).
2. Gastric erosions ( alcohol , NSIADS)
3. Oesophagitis ( 10% , usually with Hiatus hernia)
4. Esophageal varices ( 2-9%, CLD or portal v thrombosis) .
5. Mallory weiss syndrome ( esoph. Tear after severe vomiting) .
6. Vascular malformations .
7. Cancer of stomach or esophagus .
8. Aorto-duodenal fistula ( 0.2%)

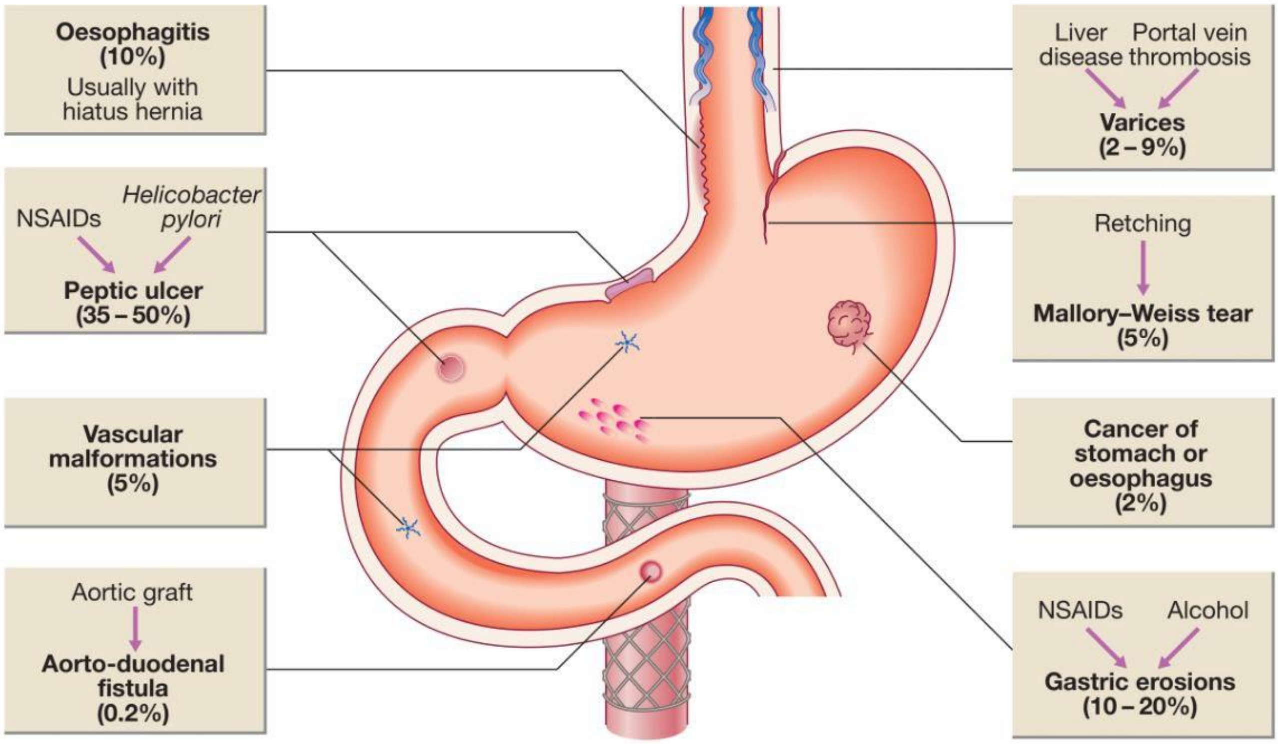



**Fig. 21.19** **Causes of acute upper gastrointestinal haemorrhage.** Frequency is given in parentheses. (NSAIDs = non-steroidal anti-inflammatory drugs)

## Clinical presentation

**1. Haematamesis :** Haematemesis is red with clots when bleeding is rapid and profuse, or black ('coffee grounds') when less severe.

**2. Melena :** is the passage of black, tarry stools containing altered blood; it is usually caused by bleeding from the upper gastrointestinal tract, although haemorrhage from the right side of the colon is occasionally responsible.

The characteristic colour and smell are the result of the action of digestive enzymes and of bacteria on haemoglobin.

**3. Hematochezia:** (red or maroon blood in the stool) is usually due to lower GI bleeding. However, it can occur with massive upper GI bleeding, which is typically associated with orthostatic hypotension.

- ***Syncope or postural dizziness*** may occur and is caused by hypotension from intravascular volume depletion. Symptoms of anaemia suggest chronic bleeding.
- ***Hemodynamic instability*** in severe cases .

## Management

1. History.
2. Examination.
3. Investigations.
4. Treatment.

*** History :**

A. symptoms? Duration , severity

B. Syncopy ?

C. PMH : previous PU? , CLD?hx of bleeding tendency or blood disorders?

D. Drugs hx.: like aspirin , warfarin , NSAIDS….

## Management

### 1. Intravenous access:

The first step is to gain intravenous access using at least one large-bore cannula.

### 2. Initial clinical assessment

- Define circulatory status.

  Severe bleeding causes tachycardia, hypotension and oliguria.

  The patient is cold and sweating, and may be agitated.

- Seek evidence of liver disease .
- Jaundice, cutaneous stigmata, hepatosplenomegaly and ascites may be present in decompensated cirrhosis.

☒ Evidence of bleeding tendency( petechi or echymosis)

☒ Examine the abdomen !

- Identify comorbidity.
- The presence of cardiorespiratory, cerebrovascular or renal disease is important, both because these may be worsened by acute bleeding and because they increase the hazards of endoscopy and surgical operations.
- These factors can be combined using the Blatchford score which can be calculated at the bedside.
- A score of 2 or less is associated with a good prognosis, while progressively higher scores are associated with poorer outcomes.

### 3. Basic investigations:

- ***Full blood count.***

  Chronic or subacute bleeding leads to anaemia but the haemoglobin concentration may be normal after sudden, major bleeding until haemodilution occurs.

  Thrombocytopenia may be a clue to the presence of hypersplenism in chronic liver disease
- ***Urea and electrolytes***.

  This test may show evidence of renal failure.

  The blood urea rises as the absorbed products of luminal blood are metabolised by the liver; an elevated blood urea with normal creatinine concentration implies severe bleeding.
- ***Liver function tests.***

  These may show evidence of chronic liver disease.
- ***Prothrombin time***.

  Check when there is a clinical suggestion of liver disease or patients are anticoagulated.
- ***Cross-matching.***

  At least 2 units of blood should be cross-matched if a significant bleed is suspected.

### 4. Resuscitation

Crystalloid fluids should be given to raise the blood pressure, and blood should be transfused when the patient is actively bleeding with low blood pressure and tachycardia.

Comorbidities should be managed as appropriate. Patients with suspected chronic liver disease should receive broad-spectrum antibiotics.

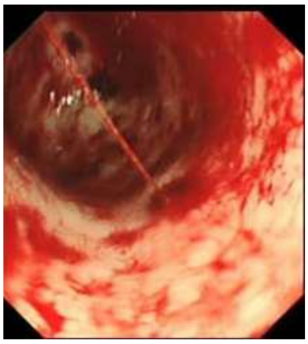
### 5. Oxygen

This should be given to all patients in shock.

### 6. Endoscopy

This should be carried out after adequate resuscitation, ideally within 24 hours, and will yield a diagnosis in 80% of cases.

Patients who are found to have major endoscopic stigmata of recent haemorrhage can be treated endoscopically using a thermal or mechanical modality, such as a 'heater probe' or endoscopic clips, combined with injection of dilute adrenaline (epinephrine) into the bleeding point ('dual therapy'). This may stop active bleeding and, combined with intravenous proton pump inhibitor (PPI) therapy, may prevent rebleeding, thus avoiding the need for surgery.



### 7. Monitoring

Patients should be closely observed, with hourly measurements of pulse, blood pressure and urine output.

### 8. Surgery

Surgery is indicated when endoscopic haemostasis fails to stop active bleeding.

### 9. Eradication

- Following treatment for ulcer bleeding, all patients should avoid non-steroidal anti-inflammatory drugs (NSAIDs) and those who test positive for H. pylori infection should receive eradication therapy , Successful eradication should be confirmed by urea breath or faecal antigen testing

**References :**

1. Davidson principles & practice of medicine .
2. Medscape .
3. Oxford handbook of medicine .